

Nikhitha Saini

AI Research Engineer | Telangana, India

Mobile: +91-7893922852 | **Email:** nikhitha.saini.162@gmail.com | **LinkedIn:** linkedin.com/in/nikhitha-saini-77 | **GitHub:** github.com/nikhithasaini58

PROFESSIONAL SUMMARY

AI Research Engineer with deep expertise in NLP and transformer architectures, fine-tuning LLMs for enterprise applications. Published internal research on model optimisation techniques. Passionate about advancing applied AI within the Indian tech ecosystem.

TECHNICAL SKILLS

Programming Languages: C++, Python, SQL, Java, R

Data Engineering: ETL, Data Pipelines, Apache Spark

NLP & Computer Vision: GPT, Computer Vision, Natural Language Processing, BERT

Soft Skills: Problem Solving, Attention to Detail, Critical Thinking, Communication

PROJECTS

Big Data Customer Segmentation Pipeline

Tech Stack: Python, ETL

- Built a Spark-based ETL pipeline processing 500GB of daily transactional data.
- Applied K-Means and DBSCAN clustering to segment 2M+ customers into 8 cohorts.
- Automated pipeline scheduling using Apache Airflow with email alerting on failure.

Predictive Churn Model for Telecom

Tech Stack: Python

- Engineered 45+ features from raw telecom usage data including RFM and behaviour signals.
- Compared XGBoost, LightGBM, and Random Forest; selected LightGBM at 88% AUC-ROC.
- Tracked 30+ experiments using MLflow and registered best model to the model registry.

Medical Image Classification using CNN

Tech Stack: Python

- Built a ResNet-50 transfer learning model for pneumonia detection from chest X-rays.
- Trained on NIH dataset (10K images), achieving 94% accuracy and 0.92 AUC.
- Implemented Grad-CAM heatmaps for model interpretability and clinical validation.

EDUCATION

B.Tech – Artificial Intelligence & Machine Learning

Jawaharlal nehru institute of engineering and technology | 2025 | CGPA: 7.37

Class XII – Telangana State board of intermediate

2021 | 89.0%

Class X – Brindavan high school

2019 | 85.0%